

Sparse Recovery and Compressed Sensing

A Model-Based Deep Learning Approach

Core question: How far can model-based structure take us — and when do we need data-driven flexibility?

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Based on: Chen, Liu, Wang, Yin — "Hyperparameter Tuning is All You Need for LISTA" • NeurIPS 2021

Seminar Roadmap

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Background

Motivation, LASSO,
ISTA / FISTA

2

Unrolling

LISTA, ALISTA,
HyperLISTA

3

Our Study

Scalar variants +
training methods

4

Results

Sparse recovery &
design space

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Images

MNIST & FashionMNIST
CS stress tests

Goal: systematically compare how much learning is needed when moving from ideal sparse signals to real compressed-sensing images.

Synthetic signals

$m=250$, $n=500$, $s=50$
Gaussian A , noiseless

Trained models

LISTA, ALISTA, HyperLISTA
+ 3 scalar designs (ours)

Image CS

MNIST & FashionMNIST pixel domain
ratios $\in \{0.25, 0.5\}$

Motivation & Problem Formulation

Sparse recovery appears everywhere:

Application	Measurements	Goal
MRI acceleration	k-space ($m \ll n$)	Image pixels
Radar	Compressed snapshots	Reflectivity
Hyperspectral	Spectral bands	Spatial cube
Compressed sensing	Random projections	Original signal

Why tractable? Natural signals have only $s \ll n$ non-zero components.

LASSO — the optimization backbone:

$$\hat{x} = \arg \min_x \frac{1}{2} \|b - Ax\|_2^2 + \lambda \|x\|_1$$

LASSO = Least Absolute Shrinkage and Selection Operator (Tibshirani, 1996)

Measurement model

$$b = Ax^* + \varepsilon$$

$A \in \mathbb{R}^{m \times n}$, $m \ll n$

x^* : s -sparse signal

ε : additive Gaussian noise

Metrics: NMSE (dB) for synthetic signals;
NMSE + PSNR + SSIM for image CS.

Classical Baselines: ISTA and FISTA

ISTA — proximal gradient descent on the LASSO objective:

$$\mathbf{x}^{(k+1)} = \mathcal{S}_{\theta} \left(\mathbf{x}^{(k)} + (1/L) A^T (b - A \mathbf{x}^{(k)}) \right), \quad L = \|A\|_2^2, \quad \theta = \lambda / L$$

FISTA — ISTA with Nesterov momentum, $O(1/k^2)$ convergence vs $O(1/k)$:

$$\mathbf{y}^{(k)} = \mathbf{x}^{(k)} + ((t_k - 1) / t_{k+1}) (\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}), \quad \mathbf{x}^{(k+1)} = \mathcal{S}_{\theta} \left(\mathbf{y}^{(k)} + (1/L) A^T (b - A \mathbf{y}^{(k)}) \right)$$

Method	NMSE @ K=16	Error energy [%]	Params
ISTA	-5.3 dB	29.5%	0
FISTA	-11.0 dB	7.9%	0

At K=16 iterations both methods are far from converged.

ISTA needs 200+ iterations for -20 dB.

Can we make 16 steps count?

ISTA = Iterative Shrinkage-Thresholding Algorithm; FISTA = Fast ISTA (Beck & Teboulle, 2009)

Deep Unrolling: From Iterations to Layers

Key idea (Gregor & LeCun, 2010): unfold K ISTA iterations into a K-layer network and learn the update parameters from data.

Classical ISTA — parameters fixed

$$\mathbf{z}^{(k)} = \mathbf{x}^{(k)} + (1/L) A^T (b - A \mathbf{x}^{(k)})$$

$$\mathbf{x}^{(k+1)} = S_{\theta}(\mathbf{z}^{(k)})$$

Parameters $1/L$ and $\theta = \lambda/L$ are fixed constants.

Fixed — no learning

Unrolled LISTA — parameters learned

$$\mathbf{z}^{(k)} = W_x^{(k)} \mathbf{x}^{(k)} + W_y^{(k)} b$$

$$\mathbf{x}^{(k+1)} = S_{\theta_k}(\mathbf{z}^{(k)})$$

$W_x(k), W_y(k), \theta_k$ are learned per layer via BPTT.

~6 M parameters — learned end-to-end

MBDL principle: preserve the known algorithmic structure (gradient step + soft-threshold);
learn only what improves finite-depth reconstruction. Training minimizes $\|\mathbf{x}(K) - \mathbf{x}^*\|^2$ via BPTT.

BPTT = Backpropagation Through Time — end-to-end backpropagation through all K unrolled layers, as when training a recurrent network

The MBDL Ladder: LISTA → ALISTA → HyperLISTA

LISTA — learn all matrices per layer, ~6 M params:

$$\mathbf{x}^{(k+1)} = \mathcal{S}_{\theta_k} (\mathbf{W}_x^{(k)} \mathbf{x}^{(k)} + \mathbf{W}_y^{(k)} \mathbf{b})$$

ALISTA — $\mathbf{W}$ analytic, learn step sizes and thresholds, 32 params:

$$\mathbf{W}^* = (\mathbf{A} \mathbf{A}^T)^{-1} \mathbf{A} \quad (\text{column-normalized})$$

$$\mathbf{x}^{(k+1)} = \mathcal{S}_{\theta_k} (\mathbf{x}^{(k)} + \gamma_k \mathbf{W}^{*T} (\mathbf{b} - \mathbf{A} \mathbf{x}^{(k)}))$$

HyperLISTA — adaptive $\theta/\beta/p$ from residuals, 3 params (grid search, zero backprop):

$$\theta^{(k)} = c_1 \mu \parallel \mathbf{A}^+ (\mathbf{A} \mathbf{x}^{(k)} - \mathbf{b}) \parallel_1, \quad \beta^{(k)} = c_2 \mu \parallel \mathbf{x}^{(k)} \parallel_0$$

$$p^{(k)} = c_3 \log (\parallel \mathbf{A}^+ \mathbf{b} \parallel_1 / \parallel \mathbf{A}^+ (\mathbf{A} \mathbf{x}^{(k)} - \mathbf{b}) \parallel_1)$$

Model	Params	NMSE	Error [%]
ISTA	0	-5.3 dB	29.5%
FISTA	0	-11.0 dB	7.9%
LISTA	6 M	-20.3 dB	0.93%
ALISTA	32	-29.8 dB	0.10%
HyperLISTA	3	-61.4 dB	0.00007%

NMSE on synthetic sparse recovery ($m=250$, $n=500$, $s=50$, noiseless, $K=16$ layers)

More structure → fewer params → better performance

This is the MBDL principle in action.

Our Design Space Study: What Should We Learn?

Research question: LISTA learns 6 M parameters; HyperLISTA uses 3 scalars. What is the minimal useful set of learnable components?

The ISTA gradient-correction update has two free scalar sequences:

$$\mathbf{x}^{(k+1)} = \mathcal{S}_{\theta_k} \left(\mathbf{x}^{(k)} + \gamma_k A^T (b - A \mathbf{x}^{(k)}) \right)$$

We designed three models, each learning a different subset:

ThresholdISTA

16 params

Learns $\theta_1, \dots, \theta_K$ only

Step fixed at $\gamma = 1/L$

Init: $\theta_0 = \lambda/L$

StepISTA

16 params

Learns $\gamma_1, \dots, \gamma_K$ only

Threshold coupled: $\theta_k = \lambda \gamma_k$

Init: $\gamma_0 = 1/L$

StepThresholdISTA

32 params

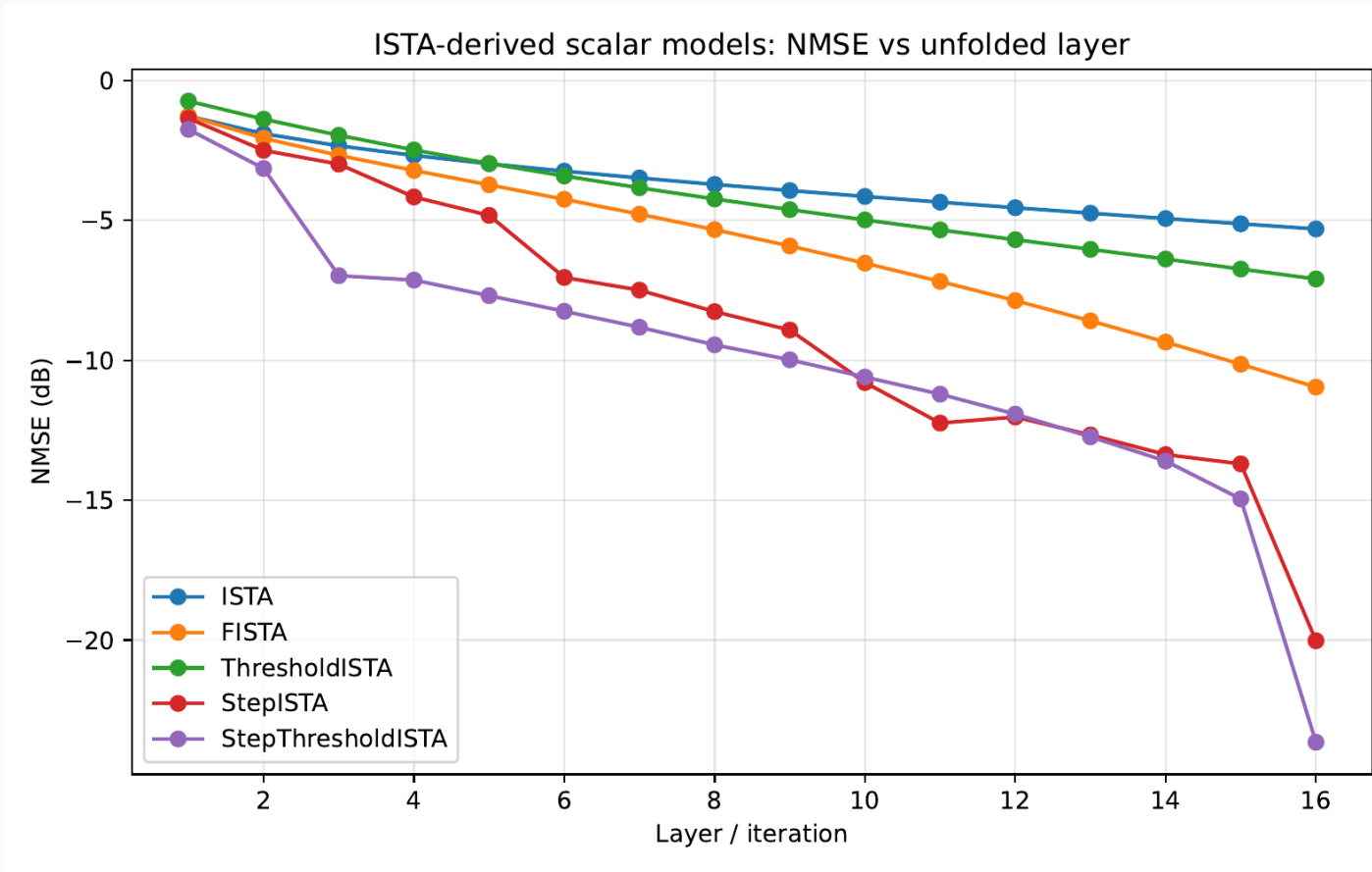
Learns both γ_k and θ_k

independently per layer

Init: ISTA values

All parameters softplus-reparameterized (positivity). All initialized to reproduce ISTA exactly before training.

Our Results: Scalar Models



Model	NMSE	Params
ISTA	-5.3 dB	0
FISTA	-11.0 dB	0
ThresholdISTA	-7.1 dB	16
StepISTA	-20.0 dB	16
StepThresholdISTA	-23.6 dB	32
LISTA-L1 (ref.)	-20.3 dB	6 M

Key finding

Learning γ_k alone (16 scalars) already matches full LISTA (6 M parameters).

Decoupling θ_k adds 3 dB at zero extra cost.

StepThresholdISTA beats LISTA-L1 with 187,000× fewer parameters.

Training Methods for Unfolded Optimizers

Architecture fixed (LISTA). Only the training strategy changes.

L1 — End-to-end

$$L = \left\| \mathbf{x}^{(K)} - \mathbf{x}^* \right\|^2$$

Supervise final output only.
Standard BPTT.

L2 — Deep Supervision

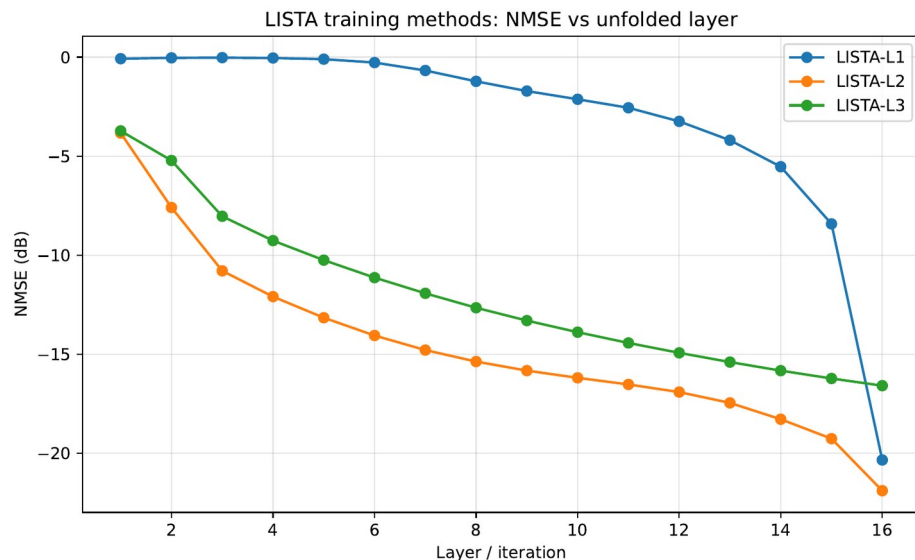
$$L = \sum_k w_k \left\| \mathbf{x}^{(k)} - \mathbf{x}^* \right\|^2$$

Supervise every
intermediate layer.

L3 — Greedy Layerwise

$$\min_{\theta_k} \left\| \mathbf{x}^{(k)} - \mathbf{x}^* \right\|^2$$

Train layer k while
layers 1...k-1 frozen.



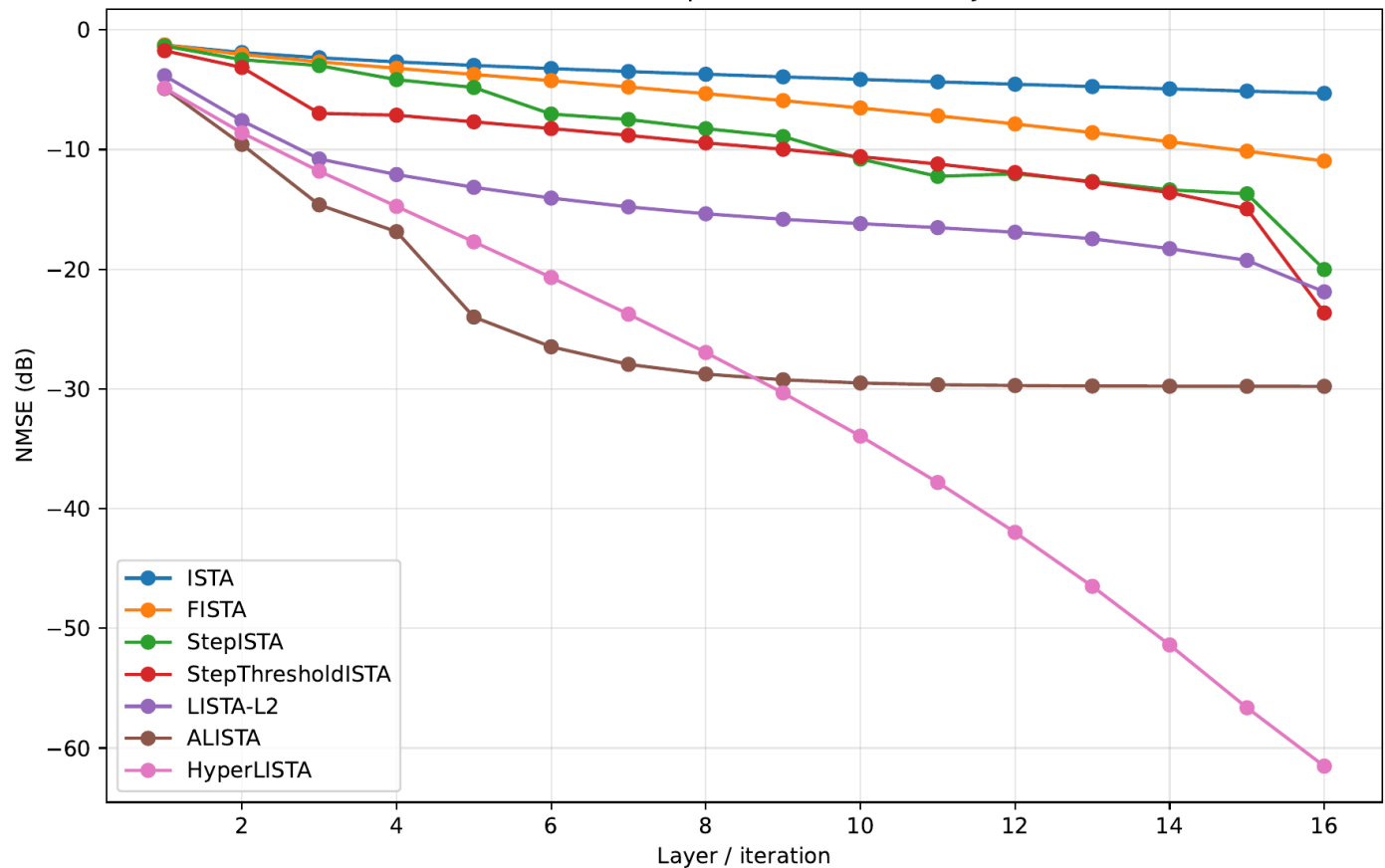
Training	NMSE @ K=16	Params
LISTA-L1	-20.3 dB	6 M
LISTA-L2	-21.9 dB	6 M
LISTA-L3	-16.6 dB	6 M
LISTA-Tied-L1	-19.1 dB	375 K

Tied = one shared (W_x, W_y, θ), RNN-style.

Implicit regularization:
-19.1 dB
with 16× fewer
parameters.

Synthetic Sparse Recovery Results

Main unfolded optimizers: NMSE vs layer

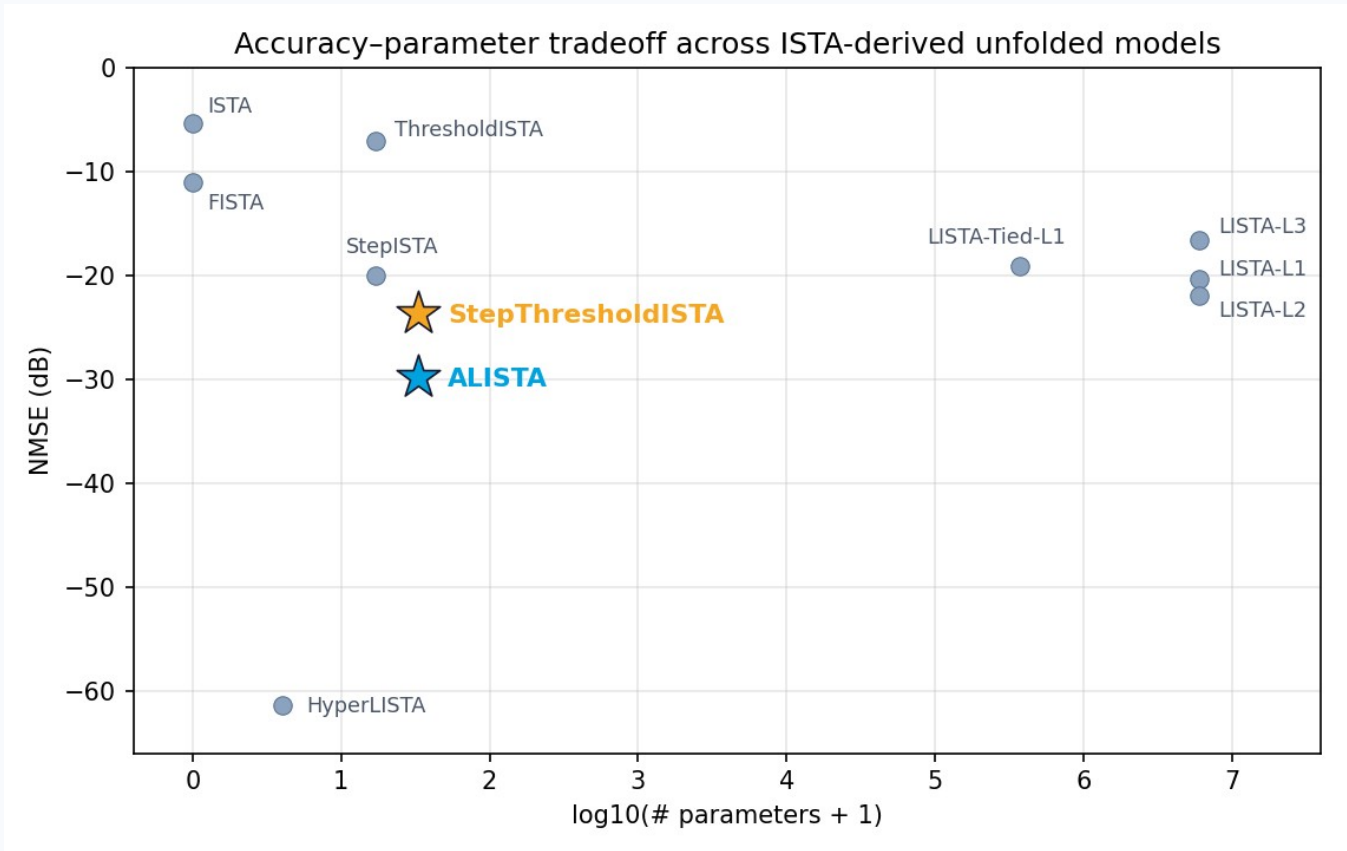


Method	NMSE	Params
ISTA	-5.3 dB	0
FISTA	-11.0 dB	0
StepISTA (ours)	-20.0 dB	16
StepThresholdISTA (ours)	-23.6 dB	32
LISTA-L2 (best LISTA)	-21.9 dB	6 M
ALISTA	-29.8 dB	32
HyperLISTA	-61.4 dB	3

Each family is shown with its best variant (LISTA → L2; full L1/L2/L3 comparison on the previous slide).

ALISTA (32 params) beats every LISTA variant; HyperLISTA (3) beats everything.

Design-Space Map



Model	Params	NMSE	What drives it
ThresholdISTA	16	-7.1 dB	learned θ_k
StepISTA	16	-20.0 dB	learned γ_k
StepThresholdISTA	32	-23.6 dB	$\gamma_k + \theta_k$
LISTA-L2	6 M	-21.9 dB	full matrices + L2
ALISTA	32	-29.8 dB	analytic W
HyperLISTA	3	-61.4 dB	adaptive residuals

StepThresholdISTA (32 scalars)
beats even the best LISTA (L2, 6 M params).
The step-size schedule γ_k is the bottleneck.

Real Image CS: FashionMNIST Pixel Domain

Setup: flatten 28×28 image to $x \in [0,1]^{784}$, measure $b = Ax$, ratio $m/784 \in \{0.25, 0.5\}$. FashionMNIST: ~51.5% of pixels exactly zero.

Method	NMSE	PSNR	SSIM	Params
ISTA	−1.2 dB	8.9	0.13	0
FISTA	−1.2 dB	8.9	0.14	0
StepThresholdISTA	−1.2 dB	8.9	0.13	32
ALISTA	−0.9 dB	8.4	0.12	32
HyperLISTA	−0.3 dB	7.7	0.09	3
LISTA-L2	−13.7 dB	22.3	0.82	12 M

ratio = 0.25

Signal model mismatch

All structured methods assume i.i.d.

Gaussian sparse signals.

FashionMNIST pixels are bounded $[0,1]$, spatially correlated, and only soft-sparse.

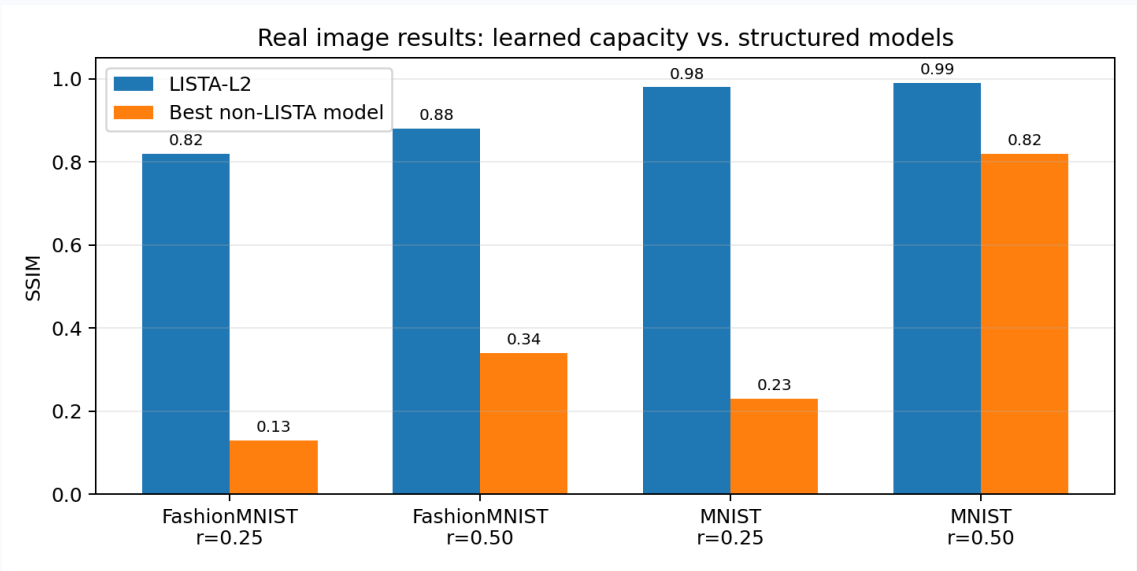
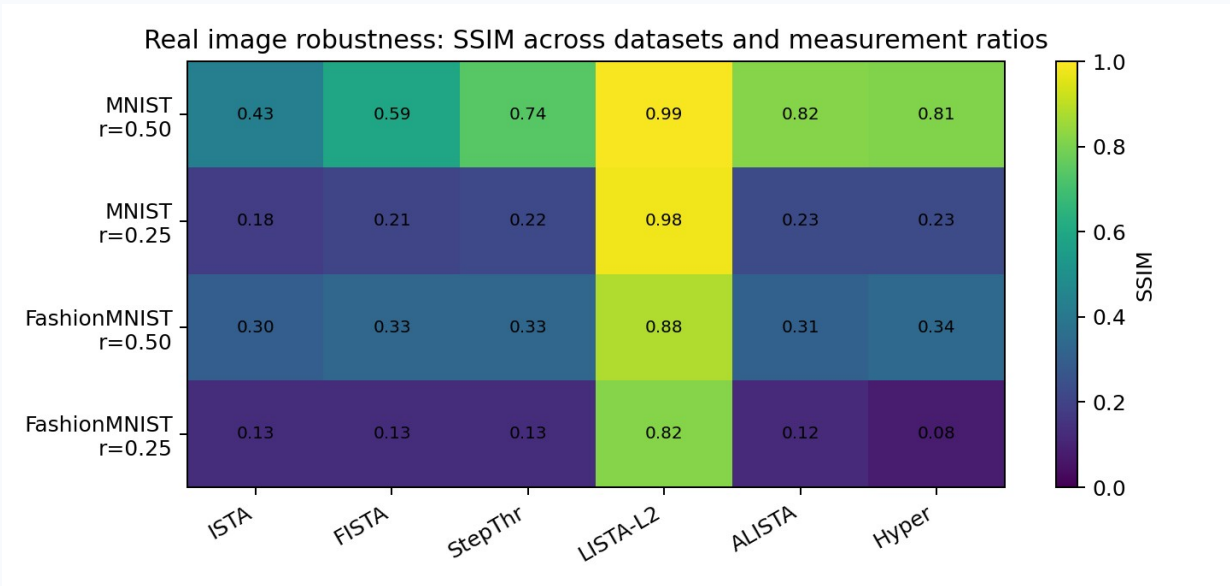
Why LISTA wins

LISTA-L2 is assumption-free —
it learns the actual image distribution
from 51 K training images.

Wrong prior is worse than no prior.

Robustness Across Datasets and Measurement Ratios

SSIM across {MNIST, FashionMNIST} × ratio {0.25, 0.5}: harder data and fewer measurements widen the gap.



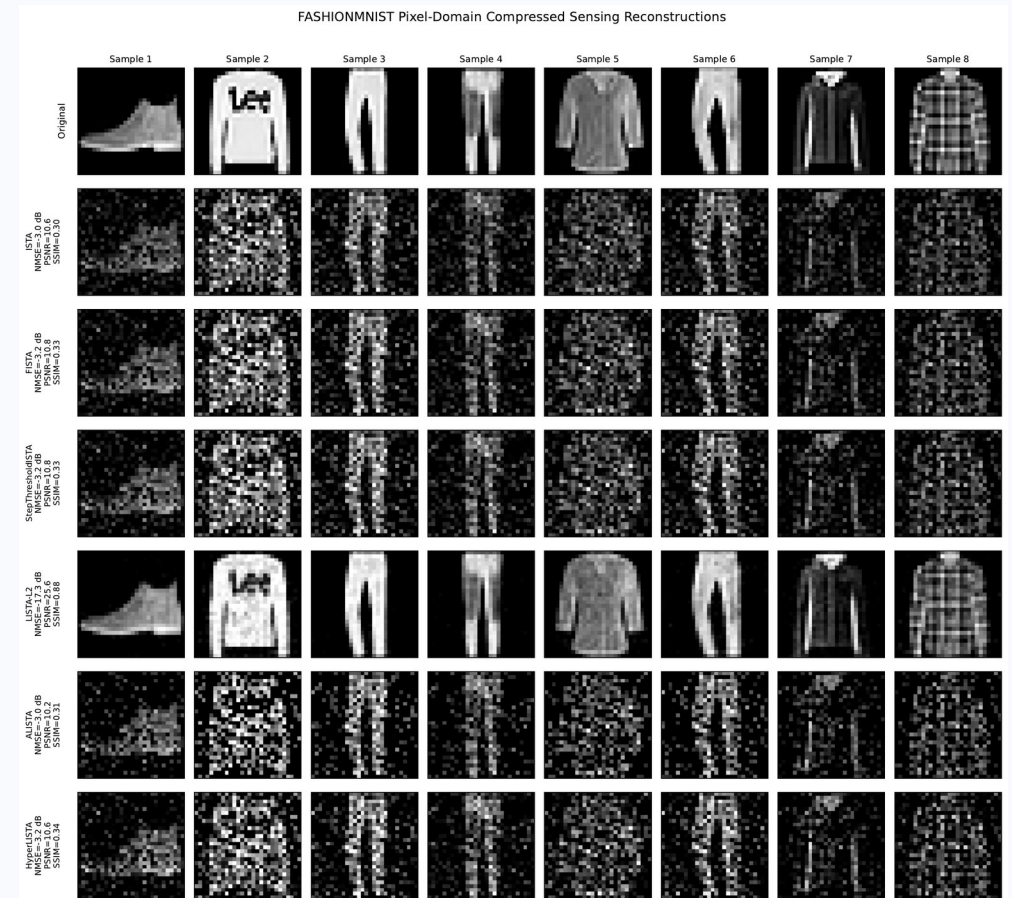
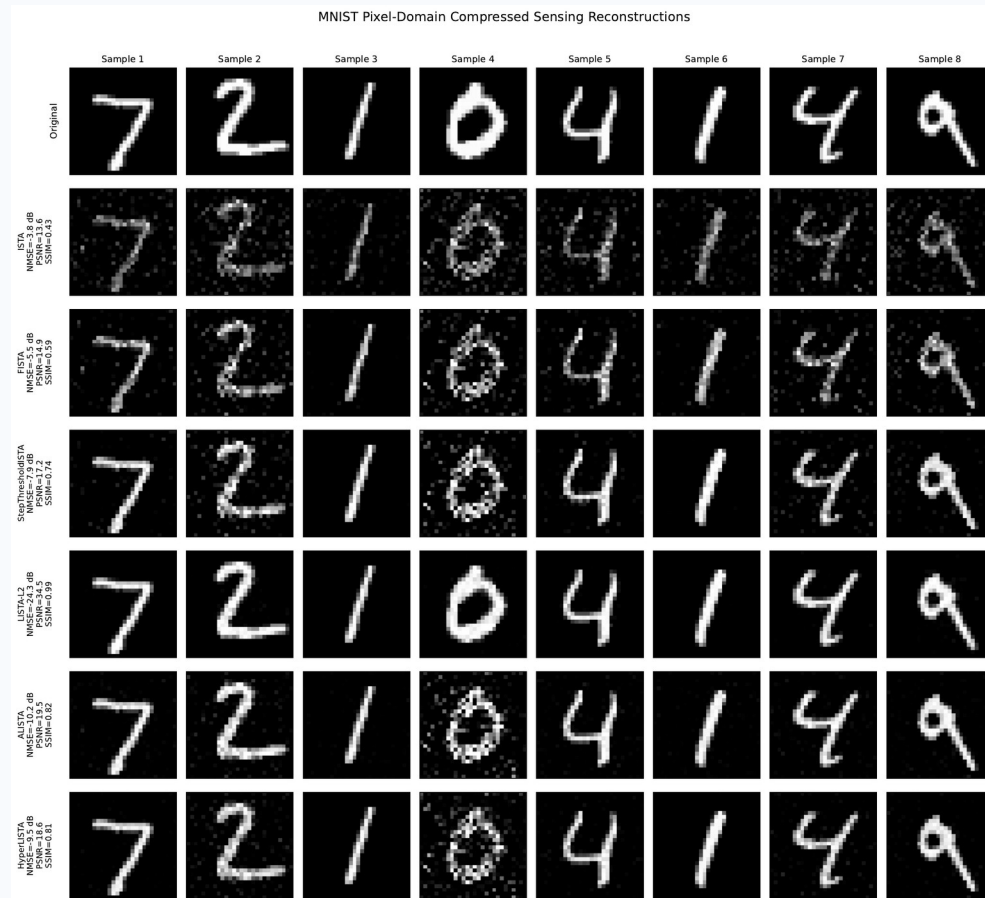
Trend

MNIST @ 0.5 is easy for everyone; FashionMNIST @ 0.25 breaks every structured model. LISTA-L2 stays ≥ 0.82 SSIM everywhere.

Interpretation

The sparse-Gaussian prior degrades gracefully on MNIST (~78% zeros) but fails on FashionMNIST (~51% zeros, textured).

Visual Stress Tests (ratio = 0.25)



LISTA-L2 recovers clean structure on both datasets. ALISTA / HyperLISTA zero out valid signal on FashionMNIST — the Gaussian-sparse threshold is miscalibrated for textured images. Wrong prior is worse than no prior.

Conclusions

1 Structure wins when assumptions match.

HyperLISTA's 3 scalars beat 6 M learned parameters by 41 dB on synthetic sparse data.

2 Step sizes matter most.

StepISTA (16 scalars) matches LISTA. Decoupling θ_k adds 3 dB. Matrix learning is largely redundant.

3 Weight tying is implicit regularization.

LISTA-Tied matches LISTA-Untied (-19.1 vs -20.3 dB) with 16× fewer parameters.

4 Wrong prior actively hurts.

ALISTA / HyperLISTA score below ISTA on real images. LISTA wins by making no structural assumptions.

MBDL hierarchy on synthetic data:

ISTA → LISTA → ALISTA → HyperLISTA

0 → 6M → 32 → 3 params

-5 → -20 → -30 → -61 dB

Future work

Apply models in DCT / wavelet domain — images are sparser there, closer to the Gaussian-sparse model.

Design image-aware thresholds for bounded, non-negative pixel structure.

Code & References

Code — public GitHub repository (notebooks 01-04 + src modules):

github.com/ThEpiCake/hyperlista-mbd

References:

- [1] A. Beck and M. Teboulle, "A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems," SIAM J. Imaging Sciences, 2009.
- [2] K. Gregor and Y. LeCun, "Learning Fast Approximations of Sparse Coding," ICML 2010.
- [3] X. Chen, J. Liu, Z. Wang, W. Yin, "Theoretical Linear Convergence of Unfolded ISTA and Its Practical Weights and Thresholds," NeurIPS 2018.
- [4] J. Liu, X. Chen, Z. Wang, W. Yin, "ALISTA: Analytic Weights Are As Good As Learned Weights in LISTA," ICLR 2019.
- [5] X. Chen, J. Liu, Z. Wang, W. Yin, "Hyperparameter Tuning is All You Need for LISTA," NeurIPS 2021.
- [6] N. Shlezinger and Y. C. Eldar, "Model-Based Deep Learning," Foundations and Trends in Signal Processing, vol. 17, no. 4, 2023.
- [7] Course lecture notes 1-8, Model-Based Deep Learning (361.2.2320), Ben-Gurion University, 2026.

Thank you very much for listening!